DP Biology – D4.1

Natural selection

- What can change allele frequencies in a population?
- What part does reproduction play in natural selection?

SL statements D4.1.1–8. Textbook pp. 779–789. Allele frequency calculations are at HL.
Image: a male lesser bird-of-paradise. Or Hiltch, CC BY 2.0.



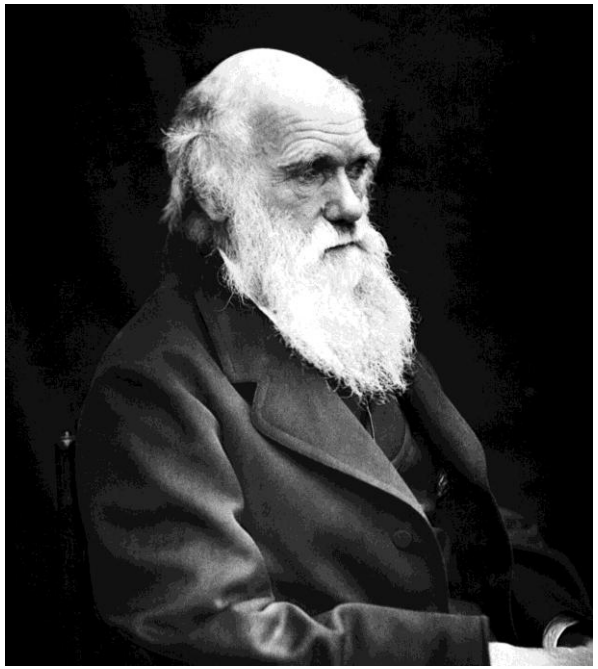
Objectives for each class

Class	Statements	By the end of the class, you can...
1	D4.1.1–4	Explain natural selection, the sources of variation, overproduction and competition, and abiotic selection pressures
2	D4.1.5–6	Use fitness to explain differences in survival and reproduction, and explain why traits must be heritable
3	D4.1.7–8	Explain sexual selection, and interpret Endler's experiments with guppies

Natural selection drives evolution

D4.1.1

Variation → Overproduction → **Competition and selection** → Survivors reproduce → Allele frequencies change



Charles Darwin. Julia Margaret Cameron, public domain.

Better-adapted individuals survive and reproduce more, passing on **heritable** traits. Over generations, **allele frequencies** change.

Big question

Why are guppies in some streams colourful and in others dull?

Sources of variation

D4.1.2

Source	What it does
Mutation	Creates new alleles: the original source
Crossing over	New allele combinations on a chromosome
Random orientation	New mixes of maternal and paternal chromosomes
Random fertilization	Combines alleles from two parents

Common mistake

Only mutation makes new alleles; sex reshuffles them. Organisms do not mutate because they need to.

Overproduction and competition

D4.1.3



Species produce more offspring than resources support. Members of a species **compete** (intraspecific competition) for food, space and mates. The largest population supported is the **carrying capacity**.

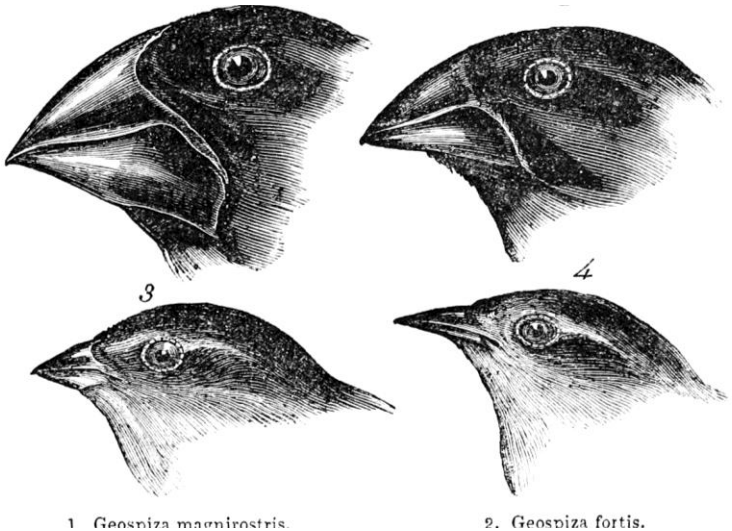
Activity: modelling beaks

Use forceps, spoons and pegs as beaks to collect beads for 30 s. Only those with enough food survive and reproduce. What does the model show, and miss?

Abiotic factors as selection pressures

D4.1.4

	Density-dependent	Density-independent
Effect	Stronger as population rises	Same at any density
Examples	Food supply, disease	Extreme temperature, drought, flood, fire



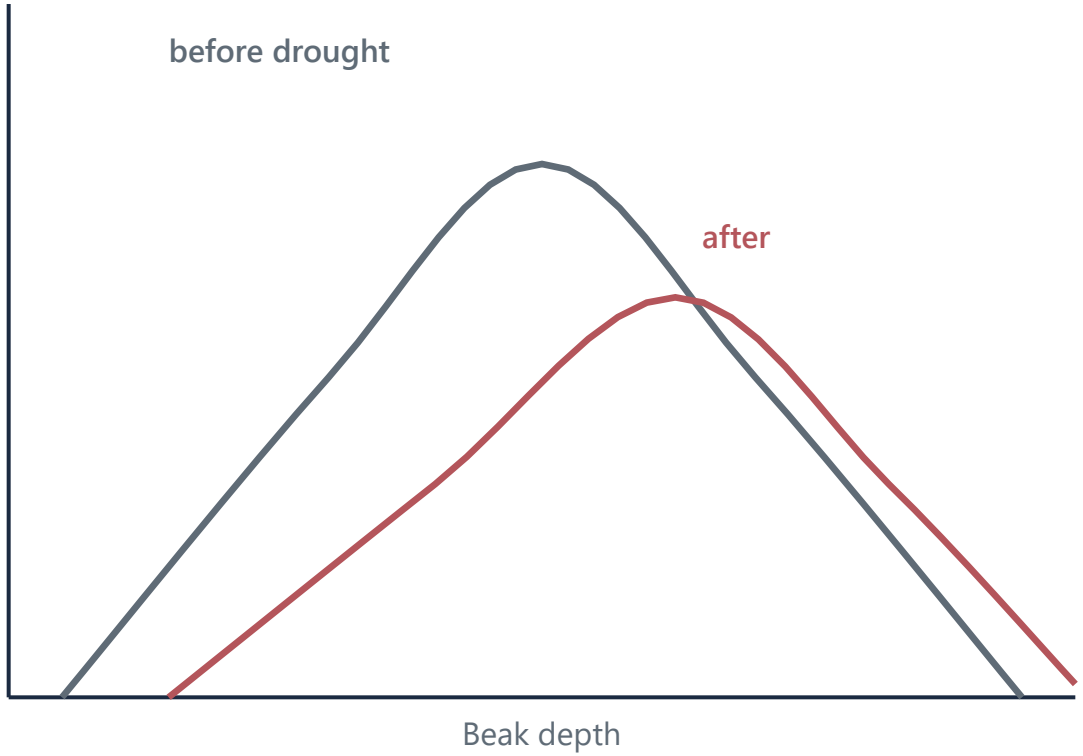
Galapagos finches. John Gould, public domain.

Example

In a Galapagos drought, finches with larger beaks survived better: they could crack the large seeds left.

Drought selects bigger beaks

D4.1.4



Sketch of the general pattern.

Read it

Mean beak depth shifted larger in one generation, because survivors had deeper beaks and passed them on.

Practice: the steps of selection

D4.1.1–4

Order

Put the five steps of natural selection in order.

Explain

Why only mutation makes new alleles.

Classify

Drought, disease, fire: density-dependent or not?

Use

variation, overproduction, competition, allele frequency

Example: the peppered moth

D4.1.4–5



Peppered moths. Chiswick Chap, CC BY-SA 3.0.

Soot darkened trees; birds spotted light moths more easily, so **dark** moths left more offspring.

Cleaner air reversed it.

Key idea

The environment did not cause the colour change: both forms existed; selection changed their frequencies.

Check-in: class 1

D4.1.1–4

1. List three sources of variation.
2. Explain why overproduction leads to competition.
3. Define carrying capacity.
4. Give an abiotic selection pressure and its effect.
5. Explain the peppered moth example.

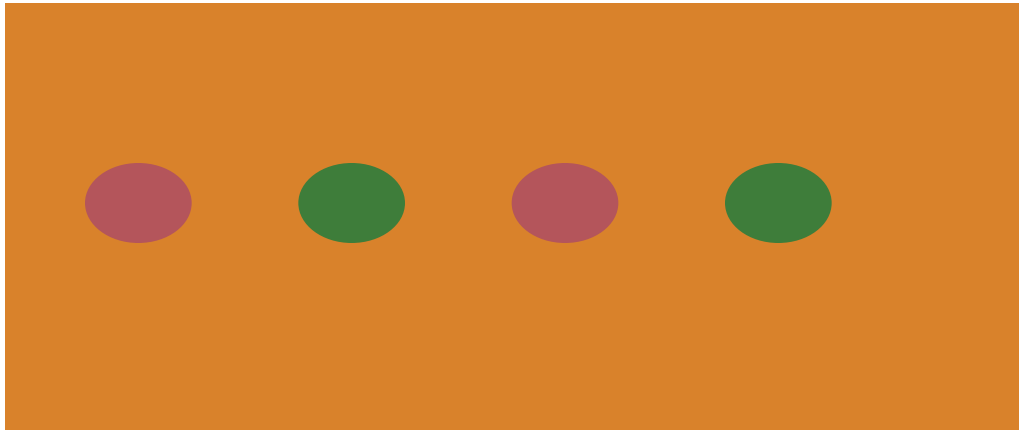
How to use this

Answer without your notes, then check against slides 3–9.

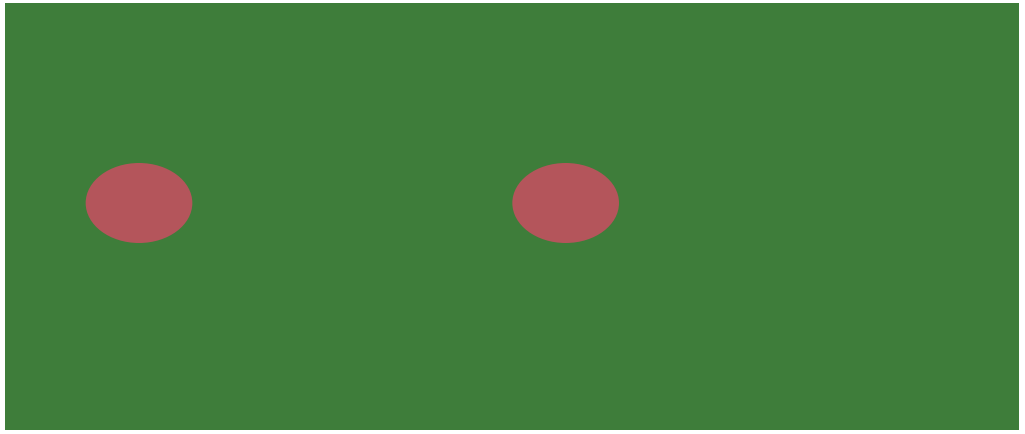
Fitness: survival and reproduction

D4.1.5

Fitness is how likely a genotype is to survive **and** leave offspring. An individual that survives without reproducing adds nothing to the next generation.



on brown soil: green beetles stand out



on leaves: brown beetles stand out

Key idea

Fitness depends on the environment, not on strength or size.

Only heritable traits can evolve

D4.1.6

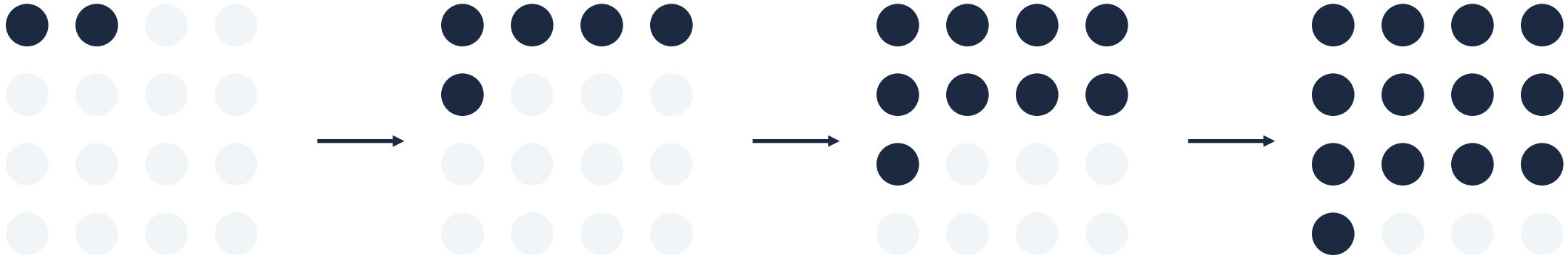
Heritable: can evolve	Acquired: cannot
Base-sequence variation	A broken leg
Migration direction in blackcaps	Trained muscles
Coat colour in mice	Pink flamingos (from food)

Three conditions

A trait evolves by natural selection only if it varies, affects survival or reproduction, and is inherited.

Watching allele frequencies change

D4.1.5–6

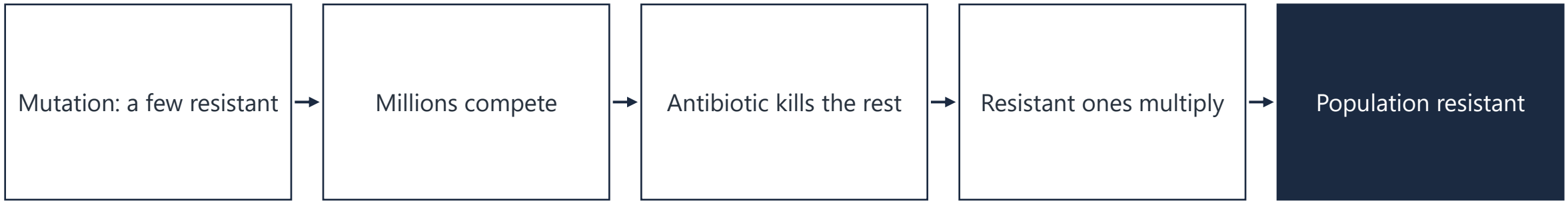


Read it

Dark moths on sooty trees: the dark allele rises from 2/16 to 13/16 in four generations. Selection changes allele frequencies; it does not create the allele.

Natural selection we can watch

D4.1.1–6



Think

Why should patients finish a course of antibiotics? Why does routine antibiotic use in farm animals worry doctors?

Practice: fitness and heritability

D4.1.5–6

Explain

Why a long-lived sterile animal has zero fitness.

Classify

Tanned skin; lactose tolerance; a scar.

Apply

Use the three conditions to explain antibiotic resistance.

Use

fitness, heritable, allele frequency, selection pressure

Check-in: class 2

D4.1.5–6

1. Define fitness.
2. Explain why fitness depends on the environment.
3. Explain why acquired traits cannot evolve.
4. State three conditions for a trait to evolve.
5. Outline how antibiotic resistance spreads.

How to use this

Answer without your notes, then check against slides 11–15.

Sexual selection

D4.1.7



Peacock. Jatin Sindhu, CC BY-SA 4.0.



Proboscis monkey. David Dennis, CC BY-SA 2.0.

Traits that help win a **mate** spread: colour, size, display, song.

Costly signals

Bright traits attract predators and cost energy. A male who can afford them may be signalling good health.

Two pressures, opposite directions

D4.1.7–8



Balance

Natural selection by predators favours dull males; sexual selection by females favours bright ones. The outcome depends on which pressure is stronger in each stream.

Modelling selection: Endler’s guppies

D4.1.8



Guppies, female and male. Amy E. Deacon et al., CC BY 4.0.

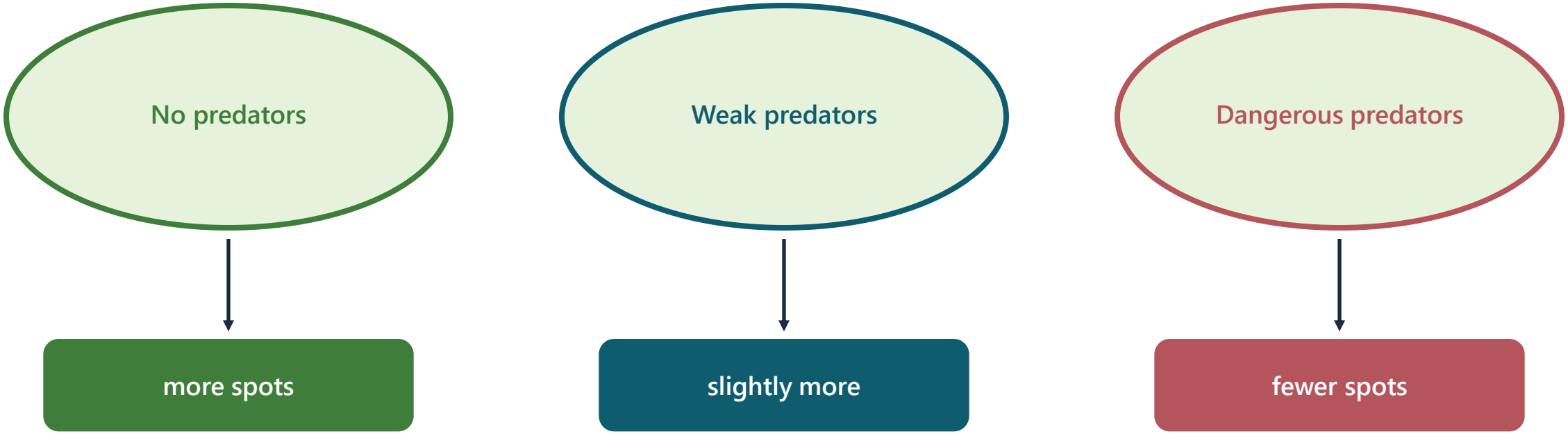
Endler bred guppies in ponds, then added **no**, **weak** or **dangerous** predators, and counted male spots over generations.

Key idea

A model with controlled pressures: traits change in only a few generations.

Endler’s design

D4.1.8



Why include weak predators?

A middle condition shows the effect is caused by predation strength, not just by the presence of any fish.

Interpreting data from a guppy experiment

D4.1.8

Pond	At start	5 months	14 months
No predators	10.0	11.5	12.5
Weak predators	10.0	11.0	12.0
Dangerous predators	10.0	9.0	8.5

Mean spots per male. Simplified from the pattern in Endler's work, for practice.

Activity

Describe each trend and explain it with natural and sexual selection. Predict what happens if males from the dangerous pond move to a predator-free pond.

Working with data: 20 generations

D4.1.8

Mean spots per male guppy. Invented values; pattern matches Endler's results.

Generation	Weak predators	Strong predators
0	11	11
5	12	9
10	13	8
20	14	7

1. Compare the two trends.
2. Explain the strong-predator trend.
3. Calculate the % change with weak predators.

Check-in: class 3

D4.1.7–8

1. Define sexual selection.
2. Explain why costly traits can be favoured.
3. Explain how predators and females pull spot number in opposite directions.
4. Outline Endler’s experiment.
5. State what the results show about the speed of evolution.

How to use this

Answer without your notes, then check against slides 17–22.

Common mistakes to avoid

Not this	But this
Bacteria become resistant because they need to.	Resistant variants exist and are selected.
The fittest are the strongest.	Fitness is survival and reproduction.
Acquired traits are inherited.	Only heritable traits pass on.
Selection creates variation.	Variation exists first.

Summary: natural selection

D4.1.1–8

	Key idea
D4.1.1	Natural selection drives evolution
D4.1.2	Variation: mutation, meiosis, fertilization
D4.1.3	Overproduction and competition
D4.1.4	Abiotic selection pressures
D4.1.5	Fitness
D4.1.6	Only heritable traits evolve
D4.1.7	Sexual selection
D4.1.8	Endler’s guppies

Check that you can...

1. **Explain** natural selection
2. **Outline** sources of variation
3. **Explain** overproduction and selection pressures
4. **Distinguish** natural and sexual selection
5. **Analyse** guppy data

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain why natural selection needs variation between individuals.
2. Give one example of a trait that is not heritable, and explain why it cannot evolve.
3. Explain why bright spots become less common in guppies when predators are present.

Image credits

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Trinidadian guppy (Poecilia reticulata) male and female.png – Amy E. Deacon, Hideyasu Shimadzu, Maria Dornelas, Indar W. R – CC BY 4.0

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